Independently routed inverse sums and common rectangle obstructions

Consolidated mathematical working draft

11 October 2026

Purpose and main conclusions

This manuscript gives a self-contained consolidation of an existing family of additive representation obstructions. It establishes one common low-rank rectangle for independently routed branches, then compares that rectangle with a Fourier matrix or a normalized complex flat unitary. There are two distinct mechanisms: componentwise projective directions over an exact field, and weighted support separators followed by a complex singular-value comparison. An earlier ordered-separation mechanism is included to make the common-rectangle and spectral statements precise and to explain which specialization the separator theorem extends.

The principal complex theorem concerns

$$M = \delta D + E + K + \sum_{i=1}^{r} L_i Q_i^{-1} R_i, \qquad d = s + 1 + \delta.$$

Here D is diagonal, E has at most s nonzeros in every row, $\operatorname{rank} K \le t$, and $\delta \in \{0, 1\}$. Each Q_i is invertible, its symmetric bipartite support lift has marked weighted half-separators of size at most q, and each endpoint has permutation-contained support, possibly with zero weights. Every branch may have unrelated left and right routes. For a complex unitary U with $|U_{ab}| = N^{-1/2}$, exact equality $U = M$ implies

$$\boxed{N \le d^2(qr + t)9^r + dq(3^r - 1) + 2(d - 1).} \tag{1}$$

The theorem requires integers $q, r \ge 1$ and $s, t \ge 0$. Invertible column-two denominators permit $q = 2$. With fixed q, s, t, (1) forces $r = \Omega(\log N)$ in any exact representation family.

A separately scoped algebraic theorem applies to the unnormalized Fourier matrix $F_{ab} = \omega^{ab}$ over a field containing a root ω of exact order N. For invertible column-two denominators and a row-degree-δ_0 extra summand S,

$$F = S + \sum_{i=1}^{r} L_i Q_i^{-1} R_i \quad \Longrightarrow \quad \boxed{N \le \delta_0(r + 1) + r(r + 1)^r.} \tag{2}$$

This theorem does not carry an uncharged low-rank correction, and it uses neither positivity nor arbitrary-row full spark. It applies at composite lengths and in positive characteristic whenever the required root exists.

All results here are restricted additive representation statements. The paper does not establish an unrestricted Fourier arithmetic lower bound, a product-grammar obstruction, an optimal constant, a practical speedup, or worldwide priority. The included results were already accepted within the project's stated proof scopes. This consolidated presentation is a working draft awaiting its own independent

review; it is not a publication-acceptance claim or a new result count. Exact historical priority remains open.

Definitions and quantifiers

Coordinates and endpoints

Write $[N] = \{0, \dots, N-1\}$, where $N \geq 1$. Submatrices use their displayed row and column sets. Complements are taken in the full coordinate set of the matrix in question. All ranks in the algebraic statements are over the stipulated field.

A matrix has **permutation-contained support** if its support is contained in a permutation matrix. Equivalently, it is a diagonal matrix times a permutation matrix; the diagonal is allowed to be singular. For a routed branch $B_i = L_i Q_i^{-1} R_i$, choose permutations p_i, s_i and endpoint weights so that

$$B_i[a, b] = \ell_{i,a} \rho_{i,b} (Q_i^{-1})[p_i(a), s_i(b)]. \tag{3}$$

For definiteness, $L_i[a, p_i(a)] = \ell_{i,a}$ and $R_i[s_i(b), b] = \rho_{i,b}$. This entrywise convention avoids a hidden transpose convention. Zero entries of a partial permutation support can be completed arbitrarily to a permutation and assigned zero weight.

The $2r$ permutations in (3) are independent. They need not match within a branch or across branches. Denominators, endpoint values, and routes may all depend on N. Coefficient magnitudes and condition numbers are unrestricted. Invertibility is required of each denominator, but not of its principal blocks or its endpoints.

Exact fields and normalized targets

In the exact-field theorem, $\mathbb{F}$ is any field and $\omega \in \mathbb{F}^\times$ has exact multiplicative order N. Thus the powers $1, \omega, \dots, \omega^{N-1}$ are distinct. If the characteristic is positive, it cannot divide N. The Fourier target is $F_{ab} = \omega^{ab}$ without a normalization requiring a square root.

In the spectral theorem the field is $\mathbb{C}$. A **normalized flat unitary** means $U^*U = I$ and $|U_{ab}| = N^{-1/2}$ for every entry. This includes normalized complex Fourier matrices at every length and normalized complex Hadamard matrices. The symbol $\|\cdot\|_2$ denotes operator norm and $\|\cdot\|_F$ Frobenius norm. Neither norm statement is asserted over a finite field.

The marked separator hypothesis

For invertible Q, define

$$H_Q = \begin{pmatrix} 0 & Q \\ Q^T & 0 \end{pmatrix}, \qquad H_Q^{-1} = \begin{pmatrix} 0 & Q^{-T} \\ Q^{-1} & 0 \end{pmatrix}. \tag{4}$$

Its undirected support graph has an upper vertex for each row of Q and a lower vertex for each column of Q. The transpose is ordinary transpose; H_Q need not be Hermitian.

Here is a sufficient hypothesis with explicit quantifiers. For every equal-size pair of target sets $A, B \subseteq [N]$ used to restrict the branch, mark the lower vertices $p_i(A)$ red and the upper vertices $s_i(B)$ blue, with unit vertex weight on every mark and zero elsewhere. There must exist a vertex set X_i of size at most q whose deletion leaves every connected component with at most half the total marked weight. Red and blue marks lie in different halves even if their numeric coordinate labels agree. Mark endpoint positions even when the associated endpoint weight is zero.

It suffices to have this property for all such equal-size pairs; more minimally, it suffices at every marking actually encountered by the construction, with a valid choice at every stage. We use the all-pairs sufficient formulation in theorem statements so that the quantifier is independent of choices made during a proof. The stronger, familiar condition requiring a half-component separator for **every nonnegative vertex weighting** also suffices. Treewidth at most $q-1$ is sufficient, but is not the definition of the hypothesis.

Separators, component groupings, and orientations may be chosen afresh for each branch and each current rectangle. No common graph, common tree decomposition, or common coordinate ordering is assumed.

Correction conventions

A row-degree bound is a maximum over individual rows. A bound on average degree or total nonzeros is not the same hypothesis. In (1), E has row degree at most s; its column degrees and values are unrestricted. The optional diagonal is charged separately by δ. The rank-t correction K is not removed by sparsity deletion; it survives and is charged to the final rank budget.

In (2), S has row degree at most δ_0. A free diagonal can be included by taking $\delta_0 = 1$. This field theorem has no separate K allowance. The parameters δ_0, δ, and d must not be identified.

Common rectangles from ordered separation

This section records the existing ordered specialization. Its algebraic rectangle selection and its complex comparison have different field requirements. It is useful when the branches come with compatible low-rank separated rectangles in their own orders; the later graph theorem does not require such a global order.

Theorem O for independently ordered sums

Let $b \geq 1$ and let each ρ_i be a nonnegative integer. For $i = 1, \ldots, b$, let B_i be a matrix over a field and let $a_i, c_i : [N] \to [N]$ be bijective row and column orderings. Suppose

$$\operatorname{rank} B_i[A, B] \leq \rho_i$$

whenever the nonempty sets A, B are strictly separated in either orientation:

$$\max a_i(A) < \min c_i(B), \quad \text{or} \quad \max c_i(B) < \min a_i(A).$$

Let h be the number of distinct specified order pairs, and put

$$R_O = \sum_i \rho_i, \qquad m = \lfloor N/2^h \rfloor.$$

There exist target row and column sets of size m on which $\sum_i B_i$ has rank at most R_O. Empty sets are permitted if $m = 0$, making the conclusion vacuous.

Proof. For current row and column sets of equal size $n \geq 2$, put $u = \lfloor n/2 \rfloor$. Let the low and high u rows be chosen in the current row order, and similarly for columns. If low rows precede high columns, keep them. Otherwise, if low columns also failed to precede high rows, the inequalities would give

$$\max a(A_{\mathrm{lo}}) \geq \min c(B_{\mathrm{hi}}) > \max c(B_{\mathrm{lo}}) \geq \min a(A_{\mathrm{hi}}) > \max a(A_{\mathrm{lo}}),$$

an impossibility. The opposite orientation therefore works. Apply this once per distinct order pair. Strict separation already obtained survives taking subsets. Repeated integer halving gives $m = \lfloor N/2^h \rfloor$. Rank subadditivity proves the claim. If a stage reaches size zero or one before all pairs have been processed, the final guaranteed size is zero and the empty rectangle suffices. □

Weighted cycle Green branches satisfy the hypothesis

Let $W[v_j, v_{j+1}] = w_j$ on a specified directed Hamiltonian cycle, with indices modulo N, and let all other entries be zero. Put $\lambda = \prod_j w_j$. Then $W^N = \lambda I$, and for $\lambda \neq 1$,

$$(I - W)^{-1} = \frac{I + W + \cdots + W^{N-1}}{1 - \lambda}.$$

An entry is the product on the unique directed arc of length $0, \ldots, N-1$, divided by $1 - \lambda$. Between strictly separated row and column positions, this product factors at a cut into a row factor and a column factor. In the reverse orientation it factors across the cycle wrap. No individual edge weight is divided out. Thus zero edges are allowed and each separated rectangle has rank at most one. Independent endpoint routes transport the two orders separately; endpoint scales, including zero scales, cannot increase rank.

A direct sum of legal cycle blocks has the same property when the block orders are concatenated: at most one component can contribute a nonzero interaction crossing a separating cut. Arbitrarily many bare cycles therefore do not by themselves require the later pseudoforest extension. Attached trees and unbounded row degree are genuinely outside this immediate cycle-order argument.

Corollary O for a normalized complex target

For a normalized flat unitary and $T = \sum_i B_i$, Theorem O gives

$$U = T \implies m^2 \leq N R_O,$$

and, for $m > 0$,

$$\|U - T\|_F^2 \geq \max(0, m^2/N - R_O), \qquad \|U - T\|_2^2 \geq \max(0, m/N - R_O/m). \tag{5}$$

For $m = 0$ the lower bounds are read as zero. The rank and norm proof is given in the next section. With b rank-one branches, $h \leq b$; fixed b gives asymptotic operator-error floor at least $2^{-b/2}$. The associated logarithmic exact branch obstruction was already present in this ordered setting. The 9^r theorem below should not be presented as a uniform numerical improvement over this 2^h retention theorem on Hamiltonian specializations. We do not need the separate prime-complex full-spark refinement here.

Rank and spectral comparison lemmas

Lemma N for complementary inverse blocks

For an invertible $n \times n$ matrix A over any field and arbitrary index sets I, J,

$$\operatorname{rank} A^{-1}[I, J] = \operatorname{rank} A[J^c, I^c] + |I| + |J| - n. \tag{6}$$

Proof. Take y supported on J and set $z = A^{-1}y$. The equation $z|_I = 0$ is equivalent to z being supported on I^c and $(Az)|_{J^c} = 0$. The map $y \mapsto z$ bijects the corresponding kernels. Their domain dimensions are $|J|$ and $n - |I|$; rank-nullity gives (6). The complements are swapped. No principal-block inverse is used. □

If a vertex separator X partitions the remaining support graph of an invertible symmetric-support matrix A into component unions V_1, V_2, then $A[V_1, V_2] = 0$. The complementary block

$$A[V_2^c, V_1^c] = A[V_1 \cup X, V_2 \cup X]$$

has rank at most $2|X|$: its entries outside the zero $V_1 \times V_2$ block occupy $|X|$ extra rows or $|X|$ extra columns. Since $|V_1| + |V_2| = n - |X|$, (6) gives

$$\operatorname{rank} A^{-1}[V_1, V_2] \le |X|. \tag{7}$$

The same is true with the sides exchanged. Applying this to (4) and restricting to lower rows and upper columns proves the inverse-rectangle rank bound for Q^{-1}. Endpoint scalings preserve the upper bound even when they vanish.

Lemma S for a low-rank rectangle of a flat unitary

Let $V = U[A, B]$ with $|A| = |B| = m$. Then

$$\|V\|_F^2 = m^2/N, \qquad \|V\|_2 \le 1, \qquad \operatorname{rank} V \ge m^2/N. \tag{8}$$

The first identity uses flat entries and the second uses coordinate compression of a unitary. The last follows by summing squared singular values, each at most one.

If $W = M[A, B]$ has rank at most a nonnegative integer R, then

$$\|U - M\|_F^2 \ge \max(0, m^2/N - R). \tag{9}$$

For $m > R$,

$$\|U - M\|_2^2 \ge \max\left(0, \frac{m^2/N - R}{m - R}\right). \tag{10}$$

For completeness, the squared distance from V to any rank-at-most-R matrix is at least the sum of the squared singular values after the first R. One can see this by projecting onto the orthogonal complement of the candidate's column space: a rank-R orthogonal projector captures at most the sum of the top R squared singular values. Those top values contribute at most R by (8), proving (9). If the block's operator error is ε, its $(R+1)$st singular value is at most ε: the span of its top $R+1$ right singular vectors intersects the kernel of W nontrivially. Every later singular value is then also at most ε. There are $m - R$ such values, giving

$$m^2/N \le R + (m - R)\varepsilon^2.$$

Coordinate restriction does not increase either norm, proving (10). The weaker (5) follows from (9) and $\|V - W\|_F^2 \le m\|V - W\|_2^2$. If $R \ge m$, the Frobenius bound is zero and (10) is not used. □

Separator selection and the complex corrected theorem

Lemma C for two colors

Suppose a branch satisfies the marked separator hypothesis, and the current target rectangle has m rows and m columns with $m \ge 4q$. There are subsets of equal size

$$m' = \left\lceil \frac{m - q}{3} \right\rceil \tag{11}$$

on which that branch has rank at most q.

Proof. Mark the routed positions as in the definition and remove a half-separator X, $|X| \le q$. Each color has at least $m - q$ surviving marks. Trim each color to exactly $u = m - q$. Components have at most m retained marks, and

$$m \le 4u/3, \qquad \text{total retained marks} = 2u.$$

Group components into two sides with each side's retained weight between $2u/3$ and $4u/3$. If a component has weight at least $2u/3$, take that component for the first side; it has weight at most $4u/3$. Otherwise accumulate components until that side first reaches $2u/3$; it remains below $4u/3$. The complement is valid in either case. Place unmarked components arbitrarily so the two sides cover all vertices outside X.

If the first side has a red and b blue marks, the second has $u-a$ red and $u-b$ blue marks. At least one oriented cross pair

$$(a, u-b), \qquad (u-a, b)$$

has both counts at least $u/3$. If both failed, combining their failure disjunctions would give either a side of total weight below $2u/3$, or $a < u/3$ and $u-a < u/3$, or $b < u/3$ and $u-b < u/3$. Each is impossible. Integer counts therefore permit $\lceil u/3 \rceil$ rows and columns. They lie on opposite component unions, so (7) and endpoint rank monotonicity give the claimed rank at most q. □

The retained fraction $1/3$ is sharp for this component-count information: when u is divisible by three, three components each with $u/3$ marks of each color allow no larger oriented cross minimum. This observation does not prove the final 9^r exponent or constants optimal for matrices.

Lemma R for removing sparse corrections

Put $d = s + 1 + \delta$ and $p = \lfloor N/d \rfloor$. Choose any p target rows A_0. The union of column supports of E in these rows, together with A_0 itself if $\delta = 1$, has size at most

$$(s+\delta)p = (d-1)p.$$

Since $N - (d-1)p \geq p$, choose p columns B_0 outside that union. Both $E[A_0, B_0]$ and $\delta D[A_0, B_0]$ vanish. This uses only row degree. The sets need not be disjoint when $\delta = 0$. The low-rank term satisfies $\operatorname{rank} K[A_0, B_0] \leq t$ and is retained. These facts persist under further restrictions. □

Theorem C for correction compatible inverse sums

Assume the complex model and all hypotheses in the opening statement, including $q, r \geq 1$ and the all-pairs marked separator condition. Then (1) holds.

Proof. Begin with the rectangle in Lemma R and process the r branches in any fixed order. Let

$$m_0 = p, \qquad m_{j+1} = \left\lceil \frac{m_j - q}{3} \right\rceil. \tag{12}$$

Whenever each preceding $m_j \geq 4q$, Lemma C constructs nested target rectangles with these sizes. A rank bound established for an earlier branch persists under every subsequent restriction. On the final square, each inverse branch has rank at most q, sparse and diagonal corrections vanish, and K has rank at most t. Thus the rank budget is

$$R = qr + t. \tag{13}$$

Ceiling monotonicity yields, for every legal stage,

$$m_j \geq \frac{p}{3^j} - \frac{q}{2}(1 - 3^{-j}). \tag{14}$$

Put

$$a = 3^r, \qquad c = (d-1) + \frac{dq(a-1)}{2}, \qquad B = d^2 R a^2.$$

Assume for contradiction that $N > B + 2c$. We verify legality before relying on the final recurrence. At the initial stage and, inductively, at each subsequent stage $j \le r-1$, (14) and $p \ge (N-d+1)/d$ give

$$\begin{aligned} m_j &\ge \frac{N-d+1}{d3^j} - \frac{q}{2}(1-3^{-j}) \\ &> dR3^{2r-j} - 1 - q/2 \\ &\ge 9q - 1 - q/2 \ge 4q. \end{aligned} \tag{15}$$

Here $N > B$, $d \ge 1$, $R \ge qr \ge q$, $r \ge 1$, and $2r - j \ge r + 1 \ge 2$. The strict threshold therefore guarantees every required use of Lemma C without circularity.

At the final stage, retaining the positive term in (14) gives

$$m = m_r \ge \frac{N-c}{da} > 0. \tag{16}$$

If $U = M$, (8) and (13) imply $m^2 \le NR$, and hence

$$(N-c)^2 \le NB.$$

But

$$(N-c)^2 - NB = N\big[N - (B+2c)\big] + c^2 > 0,$$

a contradiction. Expanding $B + 2c$ gives (1). □

Base specializations and quantitative form

The uncorrected case $s = t = \delta = 0$ gives

$$N \le qr9^r + q(3^r - 1). \tag{17}$$

With only an arbitrary diagonal, $s = t = 0$, $\delta = 1$,

$$N \le 4qr9^r + 2q(3^r - 1) + 2. \tag{18}$$

For example, $q = 2, r = 2, s = 1, t = 1, \delta = 1$ gives $N \le 3697$.

For approximation, exact equality is unnecessary. For any N at which (12) has $m_j \ge 4q$ before every selection, the construction produces a block with rank at most $R = qr + t$. Equations (9) and (10) therefore apply to M, with $m = m_r$. A sufficient condition for legality is violation of the exact cutoff, as proved in (15); legality may also be checked directly at smaller sizes. If a step is illegal, this construction by itself supplies no claimed nonempty rectangle or positive error floor.

With q, r, s, t fixed and $N \to \infty$, (12) gives

$$\frac{m_r}{N} \longrightarrow \alpha = \frac{1}{d3^r}.$$

Consequently, uniformly over the admissible choices of matrices and routes at each size,

$$\liminf \|U - M\|_2 \ge \sqrt{\alpha}, \qquad \liminf \frac{\|U - M\|_F}{\|U\|_F} \ge \alpha. \tag{19}$$

Here $\|U\|_F = \sqrt{N}$. Fixed branch count cannot yield vanishing error in these normalized metrics. The floors decay as r grows; they do not imply $r = \Omega(\log N)$ for an arbitrary fixed positive error tolerance.

For exact equality with fixed q, s, t, the right side of (1) is at most $Cr9^r$ for a constant C depending only on those fixed parameters. Taking logarithms and using $\log r \le r$ for $r \ge 1$ proves the stated $\Omega(\log N)$ branch lower bound. This asymptotic statement does not apply unchanged when the correction or separator budgets grow with N.

Sufficient support conditions including column two

Tree decompositions

A tree decomposition has vertex bags that cover every support edge and whose occurrences of each graph vertex form a connected subtree. If every bag has at most q vertices, assign each graph vertex's nonnegative weight to one bag containing it. A finite weighted tree has a centroid: repeatedly move toward a component of weight greater than half the total, if one exists. No traversed edge can be crossed back in that direction, so this ends. Remove the centroid bag. Every remaining graph component lies entirely on one side of the bag tree and consequently has at most half the weight. This proves the sufficient separator property, also for disconnected graphs by joining component decomposition trees without adding bag vertices. Zero total weight is trivial.

Lemma P for an invertible column two denominator

Suppose every column of an invertible $N \times N$ matrix Q has at most two actual nonzero entries. A zero column is impossible. Form its row-support multigraph: row coordinates are vertices, two-entry columns are labeled edges, and singleton columns are labeled loops. For a connected component with v vertices and e column-edges, column independence gives $e \leq v$. Both total counts are N, so equality holds componentwise. Each component is therefore unicyclic, allowing a loop or two parallel edges as its cycle.

In the bipartite lift, a two-entry column subdivides its edge; a singleton column replaces its loop by a leaf. Thus every component of the lifted graph is a tree or unicyclic. It is a pseudoforest. Unbounded row degree, many components, and singleton columns are allowed.

Weighted pseudoforests permit two separator vertices

Give a pseudoforest arbitrary nonnegative weights of total W. If $W = 0$, or all components have weight at most $W/2$, the empty separator works. Otherwise there is a unique heavy component. A heavy tree has a one-vertex centroid with component weights at most half that tree's weight, hence at most $W/2$.

For a heavy unicyclic component, let T_v be the tree attached to cycle vertex v, including v, and let its weight be w_v. If some $w_v > W/2$, add the outside weight $W - w_v$ at the root v in T_v and choose a centroid z of that augmented tree. If $z \neq v$, the part containing v can reconnect to the outside cycle, but its actual weight is at most the augmented part's weight, at most $W/2$. Every other detached part has the same weight as in the augmented tree. If $z = v$, the cycle remainder weighs at most $W - w_v < W/2$, and the detached branches satisfy the centroid bound. Other original graph components were already light.

If instead all $w_v \leq W/2$, delete any cycle vertex x. Aggregate each remaining attached tree into its cycle root, producing a weighted path. Choose a centroid y of that path. Its two path sides have aggregate weight at most half the remaining cycle-component weight, hence at most $W/2$. Branches detached at x or y have weight at most w_x or w_y, hence at most $W/2$. Thus $\{x, y\}$ suffices. If a remaining path has zero weight, any of its vertices is a valid centroid. A genuine cycle in the bipartite lift has at least four vertices; singleton-column loops have already become tree components.

Therefore Theorem C permits $q = 2$ for every invertible column-two denominator. A forest lift permits $q = 1$. Merely assuming a fixed column degree greater than two gives no bounded-separator guarantee. Weighted-permutation Green denominators $I - W$, when invertible, have at most two actual nonzeros per column and are included, with zero edge weights allowed. □

The separate exact field theorem

Lemma F for inverse row directions

Let Q be invertible and column-two over a field, and select any set J of k rows of Q^{-1}, $1 \le k \le N$. The nonzero columns of $A = Q^{-1}[J,:]$ occupy exactly k distinct projective directions with linearly independent representatives.

Proof. Delete the k column-labeled edges in J from the row-support multigraph of Lemma P. This is deletion of denominator **columns**, not denominator rows. The resulting graph is a pseudoforest, with exactly k tree components because

$$\sum_C \big(|V(C)| - |E(C)|\big) = N - (N-k) = k,$$

and each tree contributes one while each unicyclic component contributes zero. Isolated vertices count as trees. A surviving unicyclic component need not be untouched: deleting an attached-tree edge can split off a tree while retaining a smaller unicyclic component.

Put $B = Q[:, J^c]$. Its $N-k$ columns remain independent, also when restricted to their support components. A tree with v vertices has $v-1$ such columns and local left nullity one; a unicyclic component has v columns and local left nullity zero. Hence

$$W = \{x \in \mathbb{F}^{1\times N} : xB = 0\}$$

is a direct sum of k one-dimensional spaces with disjoint vertex supports. The rows of A are independent, lie in W by $Q^{-1}Q = I$, and form a basis of W.

Choose one nonzero local null vector for each tree and stack them into a matrix H. There is an invertible $k \times k$ matrix C with $A = CH$. A column corresponding to a vertex in tree u is a scalar multiple of $C[:,u]$; vertices in the unicyclic components give zero columns. Every local null vector has a nonzero coordinate, so every one of the k independent directions appears. No division by an edge weight and no characteristic restriction has been used. □

Routing and common refinement

Restrict a routed branch to any k target rows. Choose the k distinct denominator inverse rows prescribed by its left permutation, including positions with zero endpoint weight. Lemma F supplies k directions; multiplying by the diagonal endpoint may collapse some to zero but creates none. The right endpoint permutes and scales columns. Therefore the restricted target columns admit a partition into at most k classes, each of rank at most one. Absorb zero columns into any class; an all-zero branch uses one class. A zero vector is not a projective point, but does not need a separate linear-rank class.

The partitions from different branches are unrelated. Their common refinement has at most k^r cells, and on each cell the sum of the r branches has rank at most r. This is a family of common rectangles on a fixed set of rows, distinct from the one nested square selected by graph separators.

Theorem F with a row sparse summand

Let $F_{ab} = \omega^{ab}$ over a field with ω of exact order N. Suppose

$$F = S + \sum_{i=1}^{r} L_i Q_i^{-1} R_i,$$

where $r \geq 1$, every Q_i is invertible and column-two, endpoints satisfy (3), and S has row degree at most $\delta_0 \geq 0$. Then (2) holds.

Proof. If $r \geq N$, the claimed inequality is automatic. Otherwise put $k = r + 1 \leq N$ and select consecutive Fourier rows $0, \ldots, k-1$. At most $\delta_0 k$ columns have a nonzero S entry on these rows; discard them. Refine the r branch partitions on the remaining columns as above.

Any k distinct Fourier columns on the selected rows form a Vandermonde matrix with determinant

$$\prod_{u<v} (\omega^{b_v} - \omega^{b_u}) \neq 0.$$

Thus a common-refinement cell cannot contain k surviving columns, since the branch sum there has rank at most $r = k - 1$. Every cell has at most r columns, so at most rk^r columns survive. Restoring the discarded columns proves $N \leq \delta_0 k + rk^r$. □

Any consecutive row interval, including a cyclic wrap, works by nonzero column scaling. The proof does not allow arbitrary selected rows at composite length. It gives $N \leq r(r+1)^r$ without S, and $N \leq (r+1) + r(r+1)^r$ with a free diagonal. In particular two uncorrected branches are excluded for $N \geq 19$; no attainability claim is made below this cutoff. The uncorrected field bound yields $r = \Omega(\log N / \log\log N)$, weaker than the complex separator asymptotic but with a wider field scope. For fixed δ_0, the same field asymptotic follows.

If $r = 0$, the theorem's proof is not invoked: uncorrected equality to the zero matrix is impossible for $N \geq 1$; with S alone, all Fourier entries are nonzero, so equality requires $\delta_0 \geq N$. A diagonal-only exact representation is possible at $N = 1$ and impossible at $N > 1$. For the complex model with other sparse or rank corrections and $r = 0$, Theorem C makes no claim; those corrections may themselves represent the target.

Grammar and quantifier concordance

Result	Algebra and target	Branch and correction contract	Guaranteed conclusion
Theorem O	Any field; arbitrary matrix sum	Specified independent row and column order pairs; separated rank budgets ρ_i	One square of side $\lfloor N/2^h \rfloor$; rank at most $\sum_i \rho_i$
Corollary O	Complex normalized flat unitary	Same ordered sum; no imported sparse-correction rule	Exact rank comparison and (5)
Theorem C	Complex normalized flat unitary	Invertible denominators with marked half-separators; independent permutation-contained endpoints; $\delta D + E + K$	Cutoff (1), with the legal-recurrence norm bounds (9)–(10)
Column-two specialization	Complex normalized flat unitary	Any invertible actual column-two denominators, including attached trees	Theorem C with $q = 2$; $q = 1$ for forest lifts

Result	Algebra and target	Branch and correction contract	Guaranteed conclusion
Theorem F	Exact-order-root Fourier matrix over a field	Invertible column-two denominators; independent permutation-contained endpoints; row-degree-δ_0 term only	Bound (2); no singular-value or uncharged rank-correction statement

In Theorems C and F the universal quantifier is over all admissible matrices, routes, weights, and corrections at the specified size. The existential objects are the retained rectangle, separators or direction-class partitions. They may depend on the entire proposed representation. No fixed rectangle is asserted to work for all representations simultaneously.

Theorem O counts distinct specified order pairs h separately from the number of branches. Theorem C processes r inverse branches and does not silently replace r by a number of graph templates. Theorem F counts common partition cells, not graph separators. All three constructions use rank monotonicity under restriction, but their retention formulas and hypotheses are not interchangeable.

A sparse matrix is eliminated by a support argument, whereas a low-rank matrix is charged by rank subadditivity. One cannot substitute one correction convention for the other or transfer the t term in Theorem C into (2) without a separately proved statement. No such extension is asserted here.

Scaling and edge cases

Independent **target** row or column permutations preserve flat unitarity, so Theorem C applies directly to such normalized targets. Unitary diagonal phases also preserve it. Arbitrary nonunitary diagonal target scaling does not preserve the numerical normalization in (8).

The accepted scaling rule can nevertheless be stated explicitly. If the target is $A = D_L U D_R$ with invertible diagonal D_L, D_R, transform an approximant to $\widetilde{M} = D_L^{-1} M D_R^{-1}$. The endpoint grammar, row degree, rank correction, and diagonality of the optional diagonal are preserved. Exact branch-count obstructions therefore transfer. For either norm,

$$\|A - M\| \geq d_{\min} e_{\min} \, \|U - \widetilde{M}\|,$$

where $d_{\min} = \min |D_L[a,a]|$ and $e_{\min} = \min |D_R[b,b]|$. This follows by applying the norm inequality to multiplication by the inverse diagonals. Thus an absolute error floor must be multiplied by these minima. Relative operator floors can be weakened by the factor $\kappa_2(D_L)\kappa_2(D_R)$. A condition number alone does not set an absolute scale; even scalar rescaling changes it. Singular target scaling and arbitrary dense preconditioners are outside this transfer rule.

Other safeguards are worth collecting:

- Empty separators, zero graph weights, zero endpoints, and disconnected supports are permitted. The proof marks zero endpoint positions conservatively.
- Integer ceilings and floors are part of the statements. No asymptotic argument is used to justify a finite recurrence step.
- The assumption $q \geq 1$ avoids a separate zero-separator convention. The results do not assert a zero right-hand side for a zero-branch diagonal case.

- Column-two refers to actual nonzero support. Vanishing coefficients are handled before the support graph is formed. In a formal permutation-superpattern proof one must instead keep formal edges and redo the appropriate count; these are different bookkeeping methods.
- Equality at a necessary cutoff is not construction evidence. The constants for $r \geq 2$ and the separator exponent are not claimed optimal.
- A required branch count and an available branch-by-branch evaluation schedule do not prove a lower bound on every shared evaluator. Products, interleaved sums, nonmonomial endpoint mixing, and larger additive grammars remain excluded.

Relation to established literature

The following comparison uses the primary passages already identified by the bounded source audit. The references were reopened for this consolidation; this was a citation check, not a new exhaustive priority search. The proofs above are written out so the mathematical conclusions do not depend on accepting a negative literature-search claim.

Ingredient or quantifier	Established primary result	Relation to this manuscript
Swapped complementary inverse ranks	Vandebril and Van Barel, Section 3, Corollary 1	Exact classical identity behind (6); the separator rank bound is a short consequence
Rank structure induced by a graph	Chandrasekaran, Epperly, and Govindarajan, Propositions 2.1–2.3	GIRS gives inverse and sum/product rank algebra on one graph; the present selection must handle unrelated branch graphs
Tree rank profiles and algebra	Govindarajan, Chandrasekaran, and Dewilde, Theorem 4.4, Corollary 4.5, Propositions 4.6–4.8	TQS already treats tree representations and compatible same-tree algebra, including separate row and column block permutations
Weighted separator existence	Bodlaender et al., Section 2.1, Lemmas 2.1–2.2 and preceding weighted statement	Half-component separators and scalar component grouping are classical; the red/blue iteration is a specialized assembly
Componentwise projective directions	Zaslavsky, Section 2.1, contraction map and Theorem 2.1(b)	Classical frame/gain-graph antecedent of the inverse-row direction mechanism
The column-two class	Nelson and van Zwam, Section 2, frame-matrix definition and Propositions 2.1–2.2	Established matrix class; this manuscript proves its needed rank-nullspace statement directly, including characteristic two

For GIRS, the cited rank bound is expressed using border edges. A small vertex separator is not interchangeable with a small number of border edges when degrees are unbounded; (6)–(7) provide the needed vertex-based argument. Common-graph addition alone also does not select one rectangle across incompatible branch graphs. These are differences between the inspected statements and the present contract, not a proof that no derivation from GIRS is possible.

For TQS, it would be incorrect to claim that the theory allows only one simultaneous row/column permutation. The final paper permits distinct row and column block permutations. The present comparison concerns the separate branch-specific routed partitions, common-rectangle selection, and normalized target obstruction. No TQS representation problem, optimal storage theorem, or published open question is claimed solved.

The frame/gain-graph literature already supplies the structural perspective behind column-two matrices and componentwise directions. In particular, the inverse-row argument is not the invention of a new matrix class. The direct proof avoids importing a gain convention that would mishandle singleton columns over characteristic two. Consecutive-row Vandermonde nonvanishing and the singular-value energy inequalities are also standard ingredients.

The bounded audit did not locate a theorem statement with the entire independently routed, branch-specific marked-separator, sparse-plus-low-rank-correction contract and exact cutoff (1). The defensible description is a specialized synthesis/application with an explicit quantified statement. That absence of a located match is not proof of originality. Exact historical priority remains unresolved. The earlier report's temporary absence of the correction author body is now closed by the recovered matching body; this provenance correction does not settle priority.

Primary references

1. R. Vandebril and M. Van Barel, *A short note on the nullity theorem*, KU Leuven report TW397, 15 July 2004. Section 3, Theorem 1, printed p. 4, and Corollary 1, printed p. 5. Institutional text. This later primary account credits Fiedler–Markham and Gustafson; their original articles were not independently recovered in the prior bounded audit.
2. S. Chandrasekaran, E. N. Epperly, and N. Govindarajan, *Graph-Induced Rank Structures and their Representations*, arXiv:1911.05858v3. Propositions 2.1–2.3, printed pp. 5–8; high-degree Example 2.4. Primary preprint.
3. N. Govindarajan, S. Chandrasekaran, and P. Dewilde, *Tree Quasi-Separable Matrices: A Simultaneous Generalization of Sequentially and Hierarchically Semiseparable Representations*, SIAM Journal on Matrix Analysis and Applications 46(2), 1562–1586, 2025, DOI 10.1137/24M1682348. Theorem 4.4 and following algebraic results, pp. 1575–1576. Final institutional copy.
4. H. L. Bodlaender, P. G. Drange, M. S. Dregi, F. V. Fomin, D. Lokshtanov, and M. Pilipczuk, *A $c^k n$ 5-Approximation Algorithm for Treewidth*, author version dated 5 February 2016, Section 2.1, pp. 5–6. Author text.
5. T. Zaslavsky, *Biased Graphs IV: Geometrical Realizations*, Journal of Combinatorial Theory Series B 89 (2003), 231–297. Section 2.1 in the author version, PDF pp. 8–10, especially the contraction map and Theorem 2.1(b). Author text. Author-PDF page numbers are not journal page numbers.
6. P. Nelson and S. H. M. van Zwam, *On the existence of asymptotically good linear codes in minor-closed classes*, Section 2, pp. 4–5. Author text.

Theorem dependencies and proof source lineage

Item in this draft	Mathematical dependencies	Pinned source lineage
Theorem O and cycle example	Ordered half selection, arc-product factorization	ORDERED, ORDERED_REVIEW, R08
Corollary O and Lemma S	Flatness, unitary compression, singular-value rank/energy	ORDERED; CURRENT, Section 2B; R07–R08

Item in this draft	Mathematical dependencies	Pinned source lineage
Lemma N	Rank-nullity and an invertible full matrix	ROUTED_PROOF, CURRENT; classical nullity reference
Lemma C	Marked half-separator, Lemma N, two-color grouping	CURRENT, Section 2A; CURRENT_REVIEW; R07
Lemma R	Per-row support union and integer division	CORRECTION; R08, C05
Theorem C and (19)	Lemmas C, R, S; legal recurrence and threshold algebra	CORRECTION, CORRECTION_RECEIPT; R08
Lemma P and $q = 2$	Column independence; pseudoforest structure; weighted centroid	FIELD_EXTENSION; CURRENT, Section 2A; CURRENT_REVIEW
Lemma F and Theorem F	Column deletion nullspace; endpoint rank classes; Vandermonde	FIELD_BASE, FIELD_EXTENSION; R08
Scaling transfer	Invertible diagonal normalization and norm inequalities	CURRENT, Section 2B
Literature comparison	Bounded matched-statement inspection, not a proof dependency	ROUTED_PRIORITY and the linked primary texts
Current source-identity qualification	Recomputed source bytes and recovered-body identity record	READINESS, SOURCE_INDEX, CORRECTION_IDENTITY

The source named ROUTED_PROOF is the older sealed 36^r presentation. It proves the same general structural approach with a less efficient color selection. The current 9^r and $q = 2$ proofs are in CURRENT, independently reconstructed in R07 and discussed in R08. The theorem body CORRECTION gives the combined sparse/rank/diagonal formula. This draft re-exposes those existing results; it does not substitute the old bound for the current one or infer a missing strengthening from a short receipt.

CORRECTION_RECEIPT is a declared attribution-normalized export. Its recovered bytes have SHA-256 ac60fadc696be34bdea08f521b59714b2137d1a4e3874fe763578cfa062d0dfd. The recorded original canonical receipt identity is eba23a76dd87fef48f9c5006bb416ee1756745874de212d83099ab1bc6af2b74; those original receipt bytes are not present in this packet and are not reconstructed. The correction theorem body itself is canonical-byte matched. CORRECTION_IDENTITY closes provenance only and explicitly does not supply a new mathematical acceptance.

The companion SOURCE_MANIFEST.json supplies each source's full SHA-256, size, original local locator, packet copy, and role. All source copies are inert text or JSON. Mixed-topic originals are included for faithful provenance; only the sections identified by the dependency table support this manuscript. Unrelated theorems and fusion sections in those originals are outside its scope. No inherited research program was executed for this consolidation. Fresh author-side arithmetic and integrity checks, if present in the packet, are supplementary controls, not an independent proof receipt.

Frozen scope and remaining review questions

No unresolved proof gap was found while reconstructing the included statements. The final independent reviewer should still verify the exact endpoint orientation, the marked-separator quantifier, the two-color integer selection, the threshold legality, the component-nullspace count, and the target normalization directly from this manuscript.

The following remain expressly open or excluded: exact historical priority of the full contract, optimal constants and exponents, products or interleavings of sums, general shared evaluators, arbitrary arithmetic runtime, fixed-positive-tolerance logarithmic lower bounds, arbitrary nonunitary target normalization without scale factors, and extensions that change the additive grammar. No such problem is silently assumed solved by consolidating the accepted proofs.